FOOD DELIVERY APPLICATION USING DRONE TECHNOLOGY

Project Submitted to the
SRM University AP, Andhra Pradesh
for the partial fulfillment of the requirements to award the degree of

Bachelor of Technology
in
Computer Science & Engineering
School of Engineering & Sciences

submitted by

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May 2025

DECLARATION

I undersigned hereby declare that the project report **FOOD DELIVERY APPLICATION USING DRONE TECHNOLOGY** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Dr. Raja Sekhar Krovi. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

Place : .... Date : April 28, 2025

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CERTIFICATE

This is to certify that the report entitled **FOOD DELIVERY APPLICATION USING DRONE TECHNOLOGY** submitted by **VADITHE VENKATA DEVENDRA NAIK** to the SRM University-AP in partial fulfillment of the requirements for the award of the Degree of Master of Technology in in is a bonafide record of the project work carried out under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Name : Dr. Raja Sekhar Krovi	Name : Dr. Murali Krishna Enduri
Signature:	Signature: ....

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ABSTRACT

In today's fast-paced environment the demand for efficient, fast, and contactless food delivery services is rising rapidly. Traditional food delivery systems have inherent limitations including traffic jams, delivery delays, rising delivery costs, and increased environmental challenges. We propose to tackle these limitations through an innovative technology-oriented project and examine the associated potential eco-friendly and financial benefits. This project proposes to implement a Food Delivery Application using Drone technology consisting of an easy-to-use mobile application to place orders being autonomous drones that will promote quick, safe, and reliable food delivery. The customers and the drones will interact using a user-friendly mobile application where customers will be able to scroll through restaurants, place an order, pay using a secure payment gateway, and track their orders in real-time. When the customer places the order, the backend server will assign a drone, then it will optimize the flying route with GPS navigation and an AI algorithm, and then the backend server will provide the live scanning of the drone. Each drone is outfitted for the purpose of food delivery with a GPS module. The system includes real-time tracking, automated emergency protocols, geo-fenced areas, orders with payment gateways, automated fleet management of a fleet of drones with GPS-free moving autonomous drones. Our project clearly states that we have a systematic, services-based and eco-friendly and technological solution providing benefits such as reduced delivery time. Future enhancements could include AI-based supply and demand forecasting unit, charging order system using drone chargers provided at the restaurant where customers could have all orders charged together.

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Chapter 1

INTRODUCTION TO THE PROJECT

The food delivery market is rapidly incorporating new demands for quick, dependable, and contactless services. Smart cities, changing consumer behaviors, and technological advancements are among the driving forces for improvement within food delivery systems. Food delivery requires a significant amount of manual effort and has some drawbacks regarding efficiency while employing traditional delivery mechanisms, such as delivery vehicles dealing such as slow traffic and traffic jams, delays, high operational costs, and pollution. These factors exemplify the need for a food-delivery mechanism that is smarter, quicker, and sustainable, ultimately meeting customer expectations while taking the pressure off your operational reality.

The approach in this project, entitled "Food Delivery Application Using Drone Technology," improve food delivery by using drones and mobile applications that allows customers to order food. A customer can place an order through a mobile application to select available restaurants, pay securely (identifying themselves and their need), and track their order in real-time. An autonomous drone then picks up the order and delivers the food directly to the customers specified address. A circular delivery area allows customers to choose a location that is closer to a restaurant in an efficient, traffic-free network while allowing the drone to safely arrive and deliver to the customer within the delivery area, enabling deductions in drone battery life and congestion and avoiding classified aeronautical airspace.

1.1 PROBLEM STATEMENT

The traditional food delivery process has many complexities, including traffic congestion, the cost of operations, environmental pollution, and delivery inefficiencies creating delays, lost costs, and a negative experience for the customers. In addition, human error creates its own problems such as misdelivery and incorrect addresses. To address these issues there is a fantastic opportunity to use autonomous drones to ameliorate those complexities with efficiency, less pollution, and reduced operating costs. The future of rapidly making deliveries is implementing drone technologies into food delivery systems to reduce time, and carbon emissions, and decreasing operating costs so we can enhance overall efficiency of food delivery systems.[2]

1.2 OBJECTIVE :

The goal of this initiative is to create and develop a Food Delivery Application that will take advantages of drone technology to create an effective, efficient and environmentally friendly food delivery model. The first objective is to develop a mobile application that supports user capabilities like the ability for customers to place food orders, make cashless payments, and track their food delivery in real time in the application. The second objective is to design or manufacture autonomous drones that will complete food deliveries inside a circular delivery area around the restaurant or hub and will not only deliver food on time, but will use automation to schedule the flight path of the package as you area will be limited of the circular delivery area. The system will also include additional date features like GPS navigation, collision avoidance and detection and mitigation, real time track and monitoring to support the delivery of food in an efficient and safe manner. It will be promoted to use autonomous drones to do this instead of using traditional fuel powered delivery vehicles, which will lower the overall operational costs and the amount of labor by delivering food using an autonomous drone versus a traditional vehicle. Additionally, additional goals should be non environmentally impactful by using drones by delivery food versus traditional fuel powered delivery vehicles because the drones would be using an aviation fuel that is much more efficient and friendlier to carbon emissions. Lastly, the system will help to enhance the experience for customer satisfaction by sending out their food faster, consistent and contactless! [2]

1.3 SCOPE :

This document is dedicated to a Food Delivery Application which employs autonomous drones to provide quicker, more effective, and more sustainable food delivery services. The project will focus only on the implementation and development of a delivery system for drone devices, limited to a delivery area (defined as a circular area around the restaurant or hub) that follows predetermined optimized delivery timeframes and safe drones delivery. The application will have a function that allows customers to order and make payments, receive notifications, and track live delivery. The application would allow the drone fleet to delivery autonomously with GPS geo-fencing, detection of obstacles, and real-time monitoring of delivery. The current application will deliver within a defined radius but is scalable to grow and allow for future implementations to service larger delivery areas or existing urban environments as enhancements. In the future the concept of the drone based delivery model could be included in a smart city infrastructure. The current scope does not include the full implementation stage or full commercial presence as this step would only focus on testing, proof of concept, and validation of the drone based food delivery. Future initiatives might include deploying artificial intelligence (AI)-supported route optimized algorithm, advanced drone charging stations, or licensing issues associated with urban delivery (and obtaining permits).[2]

Chapter 2

MOTIVATION

2.1 MOTIVATION BEHIND CHOOSING FOOD DELIVERY APPLICATION USING DRONE TECHNOLOGY

The fast-growing food delivery industry has established a vast need for faster and more efficient solutions to satisfy consumer demand. The traditional delivery process is reliant on vehicle-based and human-based resources that encounter multiple challenges which result in poor delivery experiences or added delivery / operational costs. These challenges include traffic congestion, delays, vehicle and maintenance expenses and coercive environmental concerns. All of these structures lead to a consumer who is dissatisfied with their journeys, while also presenting increased costs to delivery agents. The endeavour will seek to mitigate these cases, by infusing drone and drone fleet technology into the food delivery system, presenting a new solution with various, varied benefits from existing logistics frameworks, procedures and infrastructures.

Drones are autonomous and are able to fly above established traffic patterns, therefore time to delivery can be substantially decreased; alternative delivery times increased, and traditional congestion delays can be bypassed. Drones begin from a cost against the food delivery agent (due to operating expense), but drones have a significantly more stable cost ratio than existing vehicle-based delivery modes of transport, meaning there is no fuel cost, human-based agent cost, furniture costs, maintenance costs and related costs. The drones are also aligned with trends that represent sustainability and reduced issues related to eco-friendly deliveries. Drones can operate issues around being zero carbon emission by only using electricity to operate when compared to the emissions released during traditional delivery options. Customers will also gain a level of tracking, reliability and real-time interaction with their deliveries on their mobile application when they know something has been dispatched directly; and their desired food is on route.

The driving motivation for this project is to use bespoke drone

Figure 2.1: Motivation.

2.1.1 Real-time problem identification and solution provision are beneficial.

The food delivery business has major real-time issues including traffic complexity, long delivery times, increased cost of service, and environmental harm. These issues lead to delivery delays, higher operational costs, and a generally inefficient system that cannot keep up with demand for faster, more reliable, and environmentally-friendly service. This project seeks to examine and mitigate the issues presented by the food delivery industry by utilizing drone technology for delivery. Drones can circumvent traffic congestion, cut down on delivery times, lower operational costs, and provide an environmentally-friendly service option through electricity use. By incorporating drones into the food delivery process, this project provides a feasible solution to the issues and compromises currently running down the efficiency of the industry, and creates a more efficient, cost-effective, and sustainable food delivery process.

Figure 2.2: Ideas to reality.

2.1.2 Selecting a variety of study subjects is beneficial.

Choosing drone technology for food delivery as an area of focus will allow for the exploration of many areas of diverse research. Researching drone technologies is similar to investigating new domains like blockchain or embedded systems; it can allow research into areas such as autonomous navigation, real-time tracking systems, routing, and/or environmental consciousness. Even if you spend time reviewing research papers and articles, journals, and/or attending workshops; you will develop an understanding of contemporary technologies in these areas and use this experience to develop better ideas and better solutions. The exploration of these areas will assist in building a strong project foundation and allow you to produce prototypes or ideas from newer technological developments. Additionally, this will help enhance your technical knowledge of a specific area and the overall quality of your project. Similarly, the exploration of these research areas will assist you in discovering new technologies or trends, and help you understand the background and review of issues and solutions regarding your project, which are essential in developing a quality and effective possible solution.

2.1.3 Understand and analyze project documentation effectively.

As part of our project, "Food Delivery Application Using Drone Technology," it is essential that we learn how to review project documentation correctly. Documented processes help us clarify the project goals, project design, project technologies used, and project workflow, while also ensuring every team member is on the same page with what we want to achieve, and keeps our development process organized. In the case of written documentation, it also helps us to track progress, ensure we identify issues before they happen, allow us to professionalize our quality of work and streamline our methods, and allow us to plan our next development evaluation or project phase. It helps create a clearer document for further improving the project and a structure that makes the overall project considered success.

2.1.4 Provides a platform for self-expression

The Food Delivery Application Using Drone Technology project provides an opportunity for self-expression aspects for us to express ourselves, show our creativity, fix what we see as challenges, and share innovative ways of doing things. The project allows room for our own original solutions. We can strategize as we want and design our solutions to accomplish our vision. So we can showcase our learning, communicate new ways of doing things, and express what we know and understand about the application of modern technology.

Chapter 3

LITERATURE SURVEY

Drones can be an effective advantage to food delivery service. Recent studies demonstrate these uses for drones in food delivery service. Bolognese's studies (2019) demonstrate that the use of drones avoid contact with vehicle or pedestrian traffic, thus saving time on delivery. Zhang et al. (2020) studied the optimization of drone flight paths to ensure that packages were effectively delivered in a timely manner. Kemp et al. (2018) created systems to ensure that drones avoid colliding with obstacles during flight and the safety of these drones will increase the trust and reliability pertaining to safety with electric craft. Davis et al. (2021) stated that drones could actually be more environmentally friendly than traditional vehicle deliveries. Finally, Smith et al. (2022) introduced the regulations that are needed to use drones commercially for delivery. Ultimately, these studies exemplify that drones can provide deliver service in a more productive, safer, and environmentally friendly manner.

3.1 TYPES

Since the concept of a systematic review was formalized (codified) in the 1970s, a basic division among types of reviews is the dichotomy of narrative reviews versus systematic reviews. The term literature review without further specification still refers (even now, by convention) to a narrative review.

The main types of narrative reviews are evaluative, exploratory, and instrumental.

A fourth type of review, the systematic review, also reviews the literature (the scientific literature), but because the term literature review conventionally refers to narrative reviews, the usage for referring to it is ”systematic review”. A systematic review is focused on a specific research question, trying to identify, appraise, select, and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.[3]

Chapter 4

DESIGN AND METHODOLOGY

4.1 WHAT IS THE ENGINEERING DESIGN PROCESS?

The Engineering Design Process is an important part of developing a system that will be robust and effective for the project we have chosen, ”Food Delivery Application Using Drone Technology.” In the first phase of engineering design, we identified the problem posed by slow or limited food delivery service and researched possible methods to use drones. Next, we put the ideas that we brainstormed into the design of your application interface and drone technology. We also created prototypes to explore the ability to provide delivery in a specific area. We then developed our drones with the help of associated company engineers. Each prototype and project was iterated, as is important when ensuring safety, speed, and customer satisfaction. Throughout the project we tested and refined all aspects of the system. An organized engineering design process allowed us to create an innovative and useful solution to a problem we resolved.[5].

4.2 RESEARCH DESIGN AND RESEARCH METHOD?

In our proposal entitled, ”Food Delivery Application Using Drone Technology,” the Research Design is the general layout that considers the way we carry out the study from beginning to end. The research design includes a processing list of the problem identification process and creation of objectives, determination of suitable technologies for drone and app integration and geographical limits for drone delivery (i.e., within a certain radius). This research design is both descriptive and exploratory. It is descriptive because it describes existing drone delivery systems; it is exploratory because it proposes New Solutions to local needs. The Research Design locks in the project to stay defined, organised and provide reliable results.

The Research Method describes the practical steps we took to obtain evidence, test concepts and confirm our system. Examples include secondary research via academic papers, industry papers, and government regulations on drone operation and a case study of existing company applications like Amazon Prime Air and various iterations of Zomato's drone usage, so that we could want apply hope to our own drone, and understand challenges and best practices, then we began this research based on how drone-based food delivery would align with user expectations, in which we contemplated our choice of survey, interviews or other means, and while we could have taken user expectations on our drone from a limited evidence and the beginnings of our project, we subsequently applied prototype development and iterative testing methods, in which we continuously refined the application and the process of operating the drone. We debated whether or not we would be utilizing these methods to ensure that our project would be based on solid evidence from real-world data, experience, and technology.

Chapter 5

IMPLEMENTATION

Developing a food delivery app begins with designing a user focused design in Figma. As a cloud-based collaborative design tool, Figma offers a robust set of tools to help with the design process, wireframing, and prototyping, guess user interfaces. Before designers can begin the design phase, they need a solid understanding of their targeted audience and the main features of the app. The key features include finding menus, placing orders, tracking deliveries and payments. Once the core features are defined and targeted audience is secure, designers will create wireframes to highlight the overall structure of the app and navigation flow. Wireframes are important for situating buttons and menus, and to visualize other features required for the usability and consistency of the app overall.

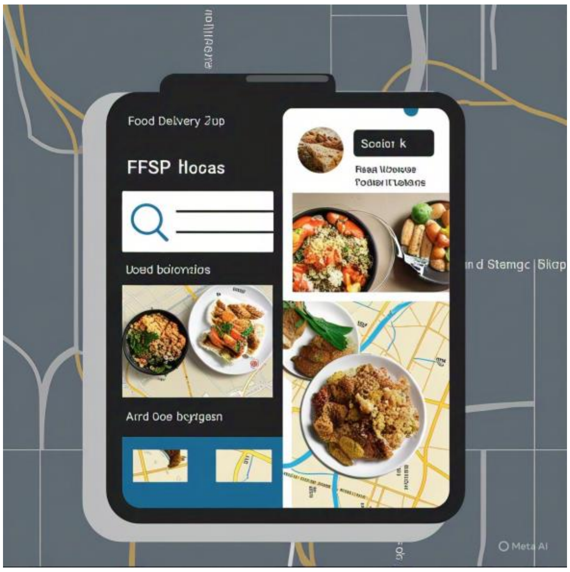

Figure 5.1: Login Page fig

During the design phase, designers will now be using Figma's vector-based editing tools to create more aspect of high quality (Apple and iOS standards) UI components. Designers will choose colors, typography, icons, etc. that suit the brand and that will resonate with their intended users. They will use the prototyping capabilities to present user interactions and provide stakeholders an opportunity to test user flows that include selecting food items, adding them to a cart, checking out, etc. Designer will collect feedback from testing sessions and ideally modify the design before a final design is developed, in order to provide the best possible user experience.

Figure 5.2: Sky High fig

When the prototype design has been finalized and approved to move to development, designers will have to handover and collaborate with developers who will have to code the Figma prototypes. The design and development collaboration happens in Figma and use Figma's prototype files. Generally, developers will create a master objects system, and then develop all the unique elements of the app based on the master. If Figma or another prototyping software was used properly, documenting reusable components should not take developers a long time. Then to save time in development, a wasteman can allow developers to export and to continue building the application with the same components unless there is feature updates from stakeholders.

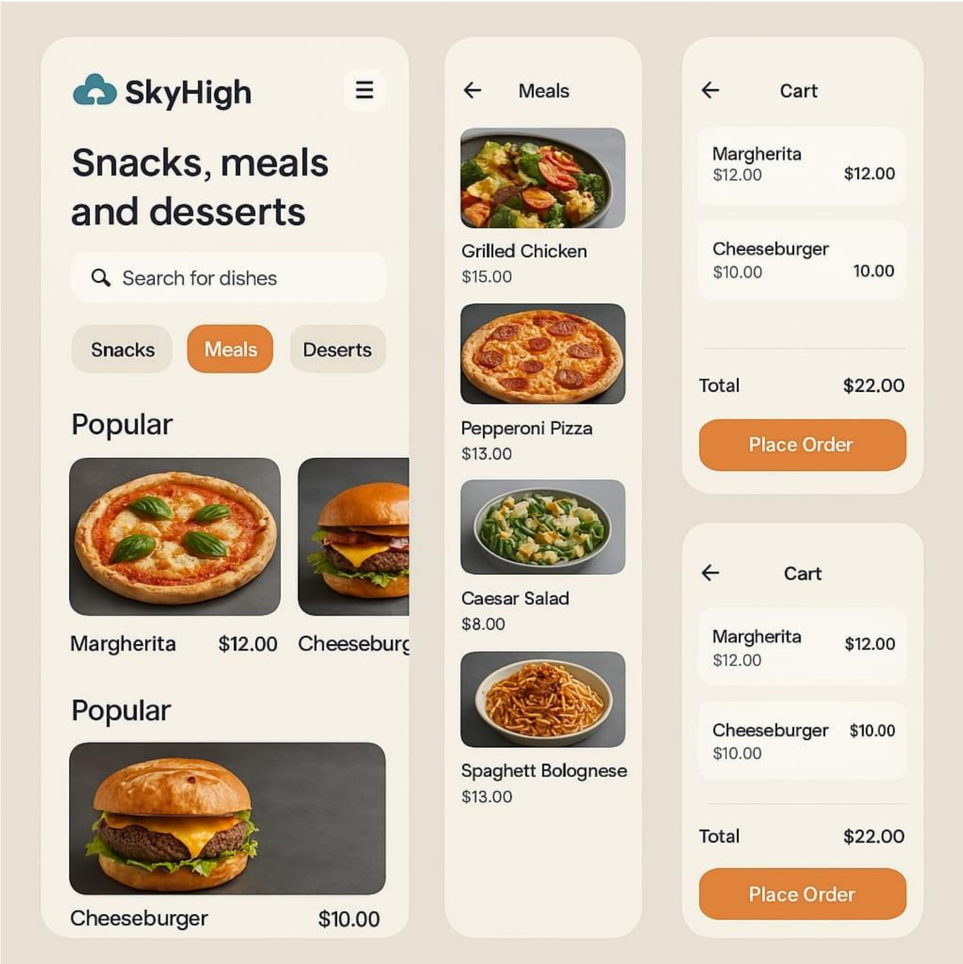

Figure 5.3: Example of a item in the above fig

Chapter 6

HARDWARE/ SOFTWARE TOOLS USED

Hardware Tools Used :

1. High-Performance Computers (Desktops/Laptops): This device is the main location where Figma and other design tools would be launching. Most of these devices will have high-resolution displays and powerful processing capabilities for rendering complex graphical interfaces and multiple design components within a single layer, all without significant latency.

2. Mobile Devices (Smartphones and Tablets): Since the app is meant for mobile device users, smartphones and tablets are key to testing and validating the design engagement responsiveness, navigation consistency, and user experience generally. In any type of design, designers are able to mimic real-world engagements through mobile devices, which allows designers to see any potential glitches in the app in real-time, as well as across different screens' dimensions and operating systems.

3. Graphics Tablets (optional): Designers can also incorporate graphics tablets (whether purpose-built like Wacom or other types) into the design workflow if they favor hand-drawn sketches, or if they need fine-grain control of digital illustrations. These devices will allow designers to be much more precise when creating or editing intricate UI components.

SOFTWARE TOOLS USED :

1. The Figma serves as a Go-Two design platform to craft the interface, manufacture interactive prototypes and cooperate in real time. Its vector editing features and cloud storage managing design assets make breeze.

With devices such as "inspection", developers can easily pull the direction -guidelines dimensions, and design files, ensuring a spontaneous transition from design to development.

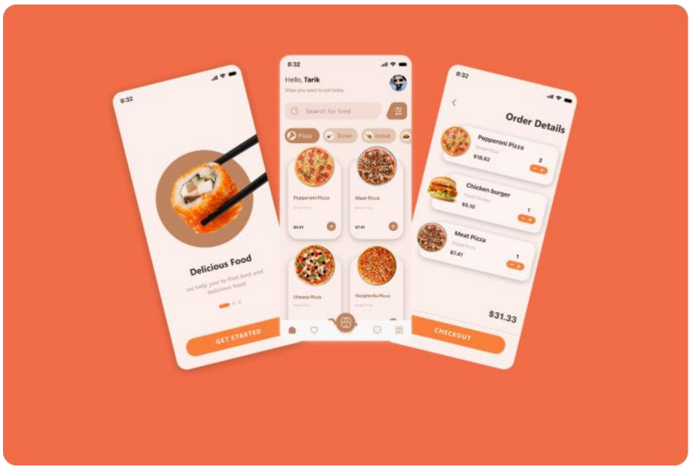

Figure 6.1: Figma

2. Prototype and purpose test equipment:

- The underlying prototypening of Figema: This feature enables the creation of interactive flows and clickable demos that mimic user interactions, making it easier to conduct a purposeful test before any coding starts.

- User Testing Platforms: Tools like Invision Freehand or Loakback are usually used to collect user feedback on prototypes, identify pain points and quickly recur on design.

3. Cooperation and Communication Equipment: - Slack, Microsoft Teams, or Zoom: These communication tools help design and development teams to maintain clear and efficient cooperation, ensuring that the response and modification is immediately addressed. - Project Management Tools (eg Tralo, Asan, or Mande.com): They keep tracks of tasks, deadlines and updates, ensuring that the entire team is align on the project milestone.

Chapter 7

RESULTS & DISCUSSION

The drones competing against traditional food delivery methods include faster delivery times and lesser environmental impact. The hurdles that lay ahead prior to the food delivery system becoming a realistic option for supply chains on a larger scale include government regulators, weather, and more reliable battery systems. AI's potential influence on optimizing a delivery route and updating the drone's navigation system to improve reliability, does offer an added reliability for utilizing drones. There is still the issue of scaling, and if technological advances continue to develop, as well as more governmental regulators climb on board, drone food delivery could possibly become standard in the urban logistics supply chain realm.

7.1 WHAT IS THE PURPOSE OF A RESULTS SECTION?

In a project, the Results section's role is to succinctly present the results, listing the data in a clear way - not to interpret (that's for the Discussion) or analyze (which is also for the Discussion). We present the collected data with tables, graphs, and charts format for clear data presentation and data management reviewing. This section will bolster the assertions made by presenting an objective description of facts supporting research questions and hypotheses. These are the key findings - the main outcomes we want to highlight so the audience readers can easily access the most important outcomes. The Results section is also the first step into the next section of Discussion where we add some context or implications to peeling back of the data. The Results section is a starting point for future research, or recommends a specific area that must be scrutinized or improved. The Results section is an avenue for reflection on the raw data to inform future conclusions later.

Figure 7.1: Final Product

Chapter 8

CONCLUSION

In summary, the potential for drone technology in the food delivery industry is significant for providing greater efficiency, shorter delivery times, and more environmentally-friendly delivery. This project has examined different aspects of drone-based delivery systems, such as the difficulty of providing autonomous navigation, improved routing, and managing margin of safety in urban settings. The use of drones for food delivery addresses inefficiencies and also provides greater efficiencies over traditional delivery methods. The introduction of drone-based food delivery reduces carbon footprints and contributes to a more sustainable better planetary health.

In this project, we have been able to apply advances in drone technology, real-time tracking, and routing methods to illustrate how service-enhancing solutions can make food delivery less cumbersome for producers and customers alike. Because of the interest in the transition to autonomous systems, the project may show the way to research and develop future autonomous systems in industries beyond food delivery.

As drone technology continues to change, the emergence of such autonomous systems in urban logistics will likely be forthcoming for the future delivery of goods and services. Assuming we continue to innovate and improve upon the regulatory landscape, it is quite possible drone-based food delivery could significantly disrupt the logistics industry.

8.1 SCOPE OF FURTHER WORK

8.1.1 Drone Efficiency Improvement:

-Enhance battery life for longer flights.

-Develop AI-powered route optimization for faster deliveries.

-Improve autonomous navigation for better safety and accuracy.

8.1.2 Service Area Expansion:

-Increase delivery range to cover larger regions or even multiple cities.

-Implement multiple drones to handle more deliveries at once.

8.1.3 Sustainability Initiatives:

-Use solar-powered drones to reduce energy consumption.

-Implement eco-friendly materials in drone construction.

8.1.4 Regulatory and Safety Compliance:

-Explore and adapt to airspace regulations for urban drone delivery.

-Integrate advanced safety features like collision avoidance and emergency landing systems.

8.1.5 Development of a project in the future :

The next steps of the Drone Technology Food Delivery Application is trying to improve drone performance, increase delivery area coverage and efficiency. Improvement can include drone battery life, AI with optimal routing and coordination of multiple drones. Additional development will prioritize safety features, with regulatory concerns and sustainability, think solar-powered drones. This project could also develop the customer experience with real-time delivery tracking features and personalized recommendations, while minimizing the operational costs to operate and offering new business models. The goal will be improved food delivery systems that are more efficient, safer, and better for the environment.

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